

STANDARD OPERATING PROCEDURE (SOP)

Ubuntu Linux Virtual Machine Setup Using VirtualBox

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Operating System: Ubuntu Linux

Virtualization Platform: Oracle VirtualBox

1. Purpose

This Standard Operating Procedure (SOP) documents the process for installing and configuring an Ubuntu Linux virtual machine using Oracle VirtualBox.

The purpose of this procedure is to provide a repeatable process for:

- Creating an Ubuntu virtual machine
 - Configuring virtual hardware
 - Installing Ubuntu Linux
 - Configuring the operating system
 - Installing required updates and software
 - Verifying the system is functioning correctly
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2. Scope

This SOP applies to the setup and initial configuration of an Ubuntu Linux virtual machine in Oracle VirtualBox.

This procedure is intended for:

- Linux system administration practice
- Virtualization practice
- Cybersecurity labs
- IT administration training

- Development and testing environments
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3. Prerequisites

Before beginning this procedure, verify that the following requirements are available.

Hardware Requirements

- Windows/macOS/Linux host computer
- Minimum 8 GB available RAM
- Minimum 30 GB available storage

Software Requirements

- Oracle VirtualBox installed
- Ubuntu ISO image downloaded
- Internet connection available

Required Files

Ubuntu ISO:

[Ubuntu ISO Download Link](#)

VirtualBox Install:

[VirtualBox Install](#)

4. Procedure

Step 1 — Install Ubuntu ISO

Objective

Install the newest version of the Ubuntu ISO for our machine.

Procedure

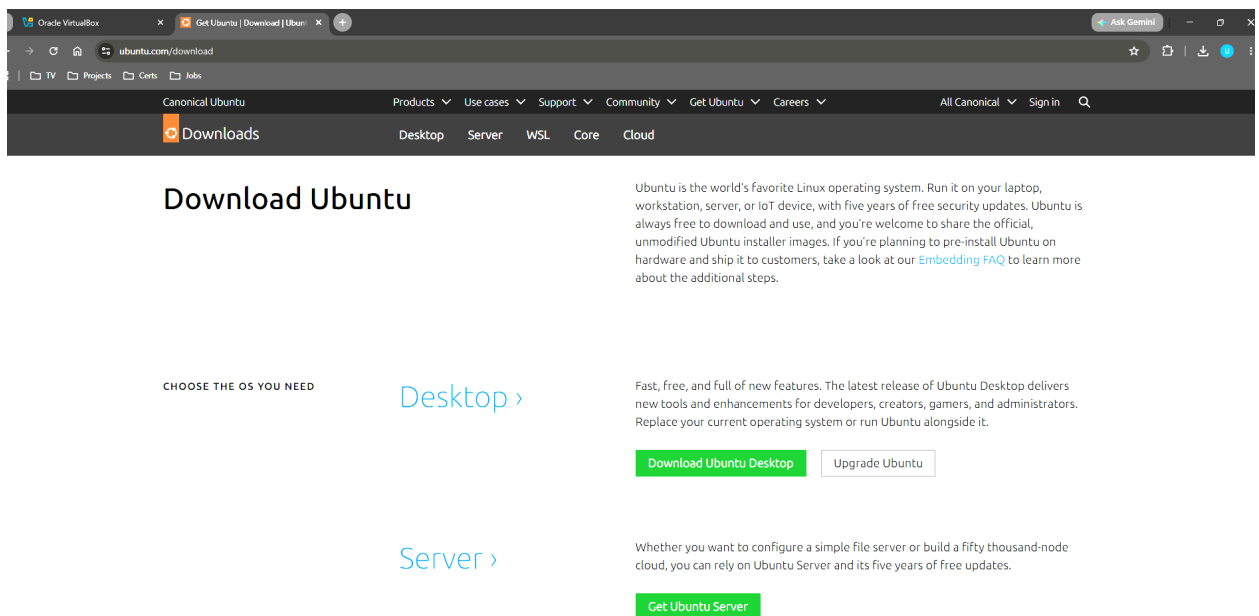
1. Visit <https://ubuntu.com/download>

For this setup, choose of these:

- **Ubuntu Desktop** if you want a GUI
 - **Ubuntu Server** if you want only command line
2. Install whichever OS version of Ubuntu you want to use (I used Desktop in this SOP)
 3. Save the ISO file in a place where you can easily access or remember where it was installed.

Screenshot

Figure 1 — Screenshot of the ubuntu ISO install page.



Notes:

Desktop is better if you are new to Linux and want a Graphical User Interface (GUI)

Server is better if you are familiar with Linux as it provides only the command line

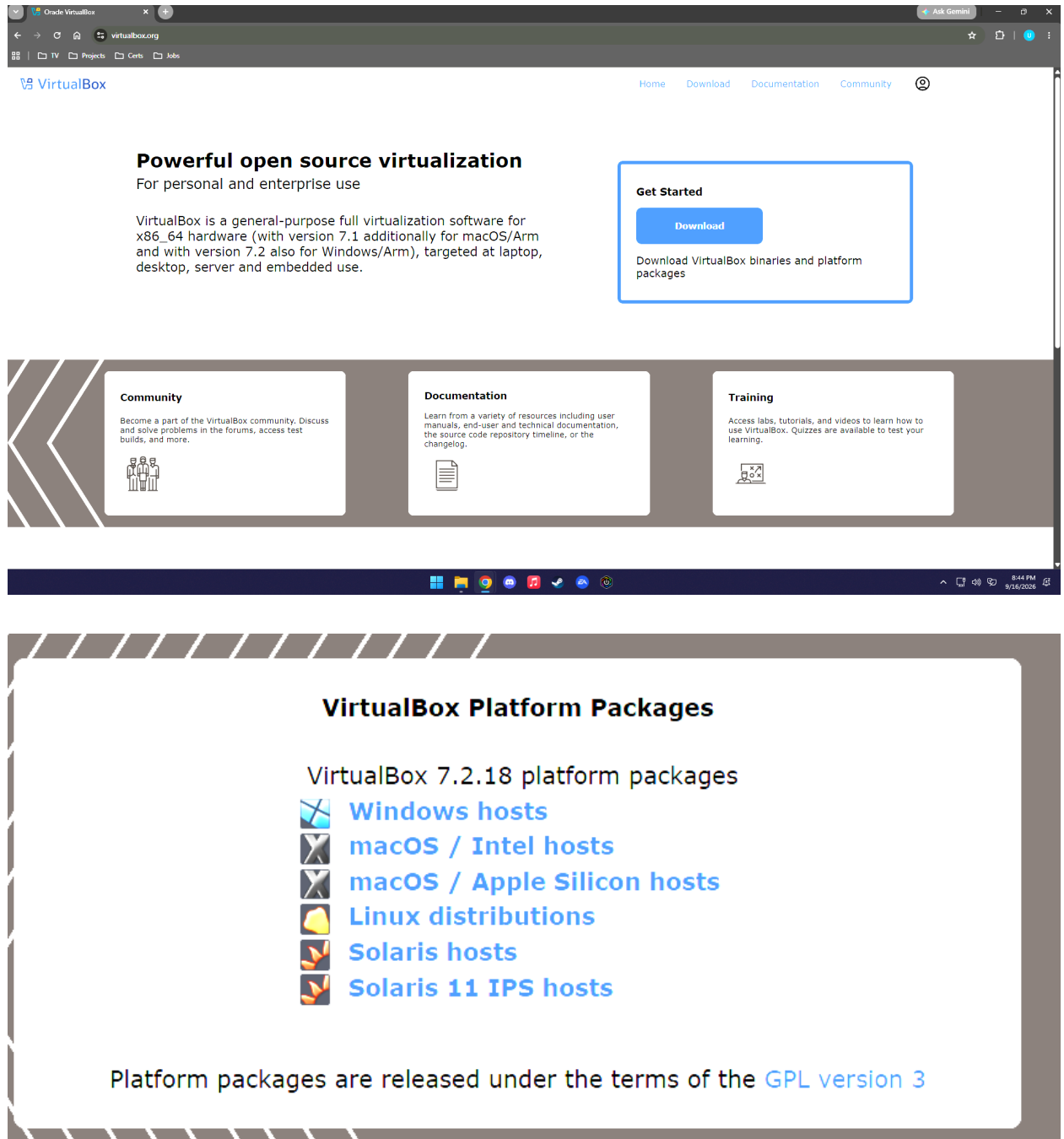
Step 2 — Installing VirtualBox

Objective

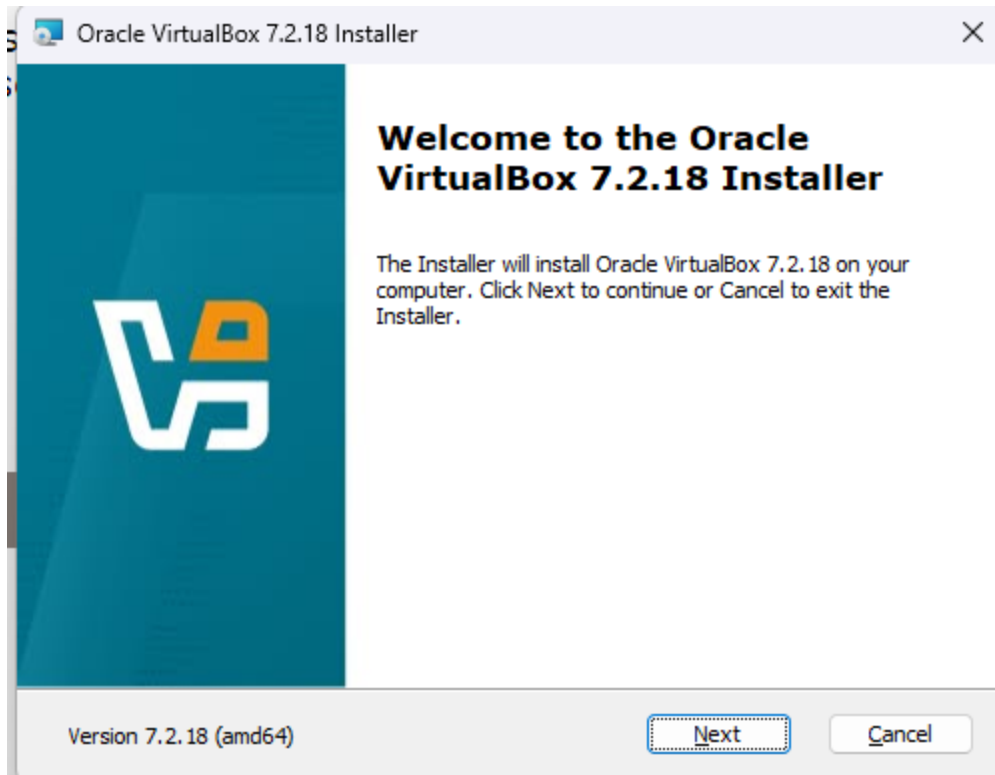
Install and configure the VirtualBox Application.

Procedure

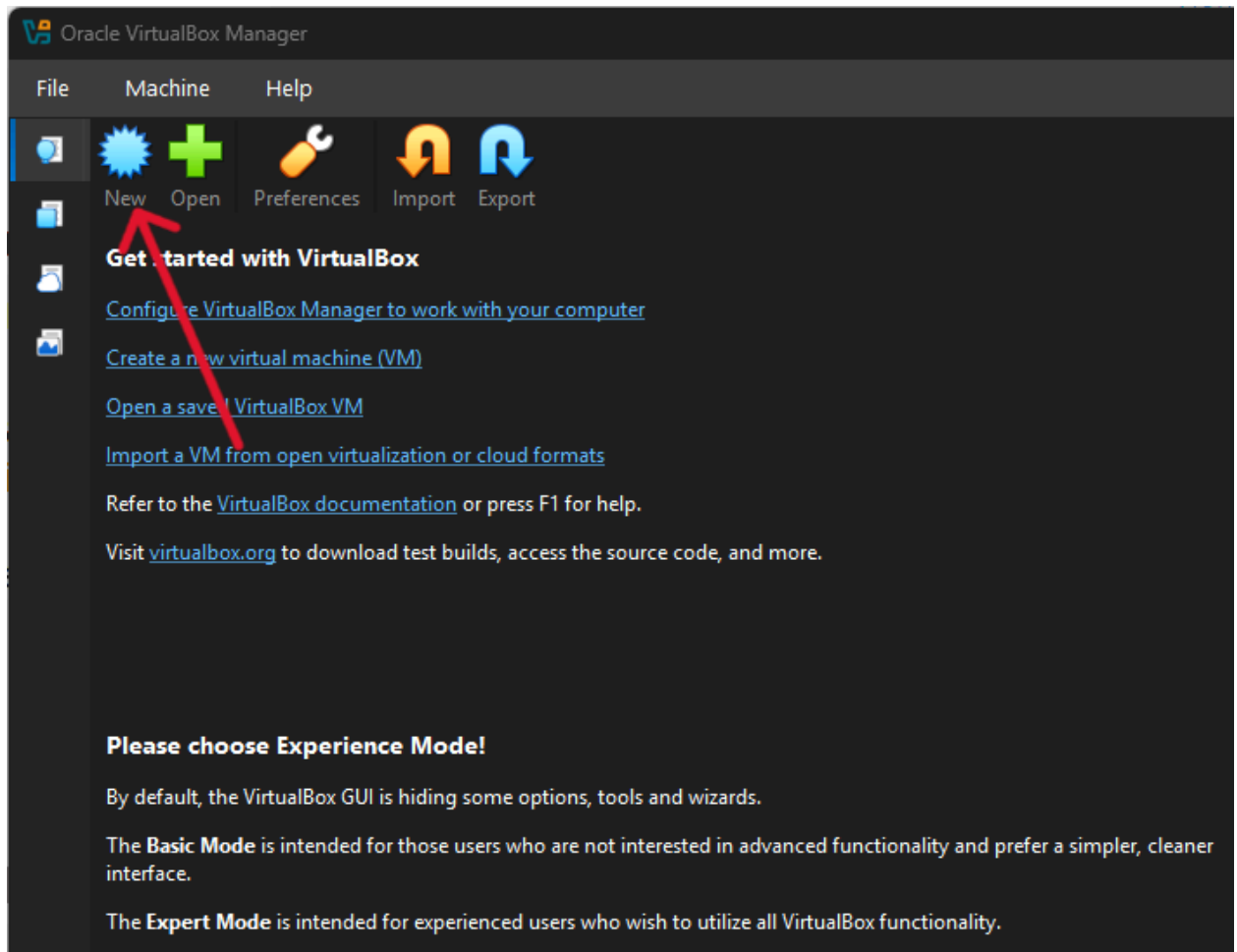
1. Visit <https://www.virtualbox.org/> to install the VirtualBox software (choose your current OS).



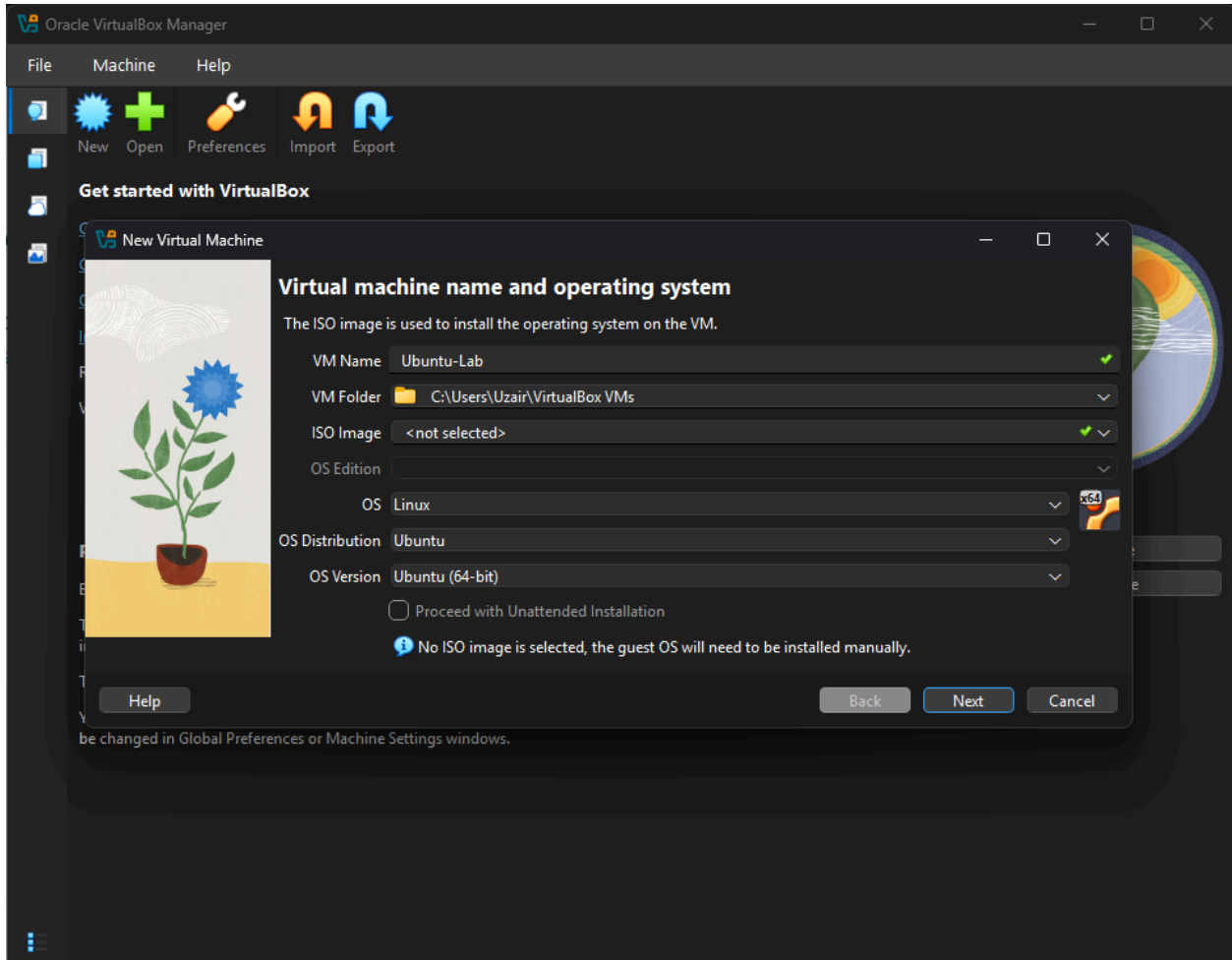
2. After download is complete, open the download file and run through the setup process.

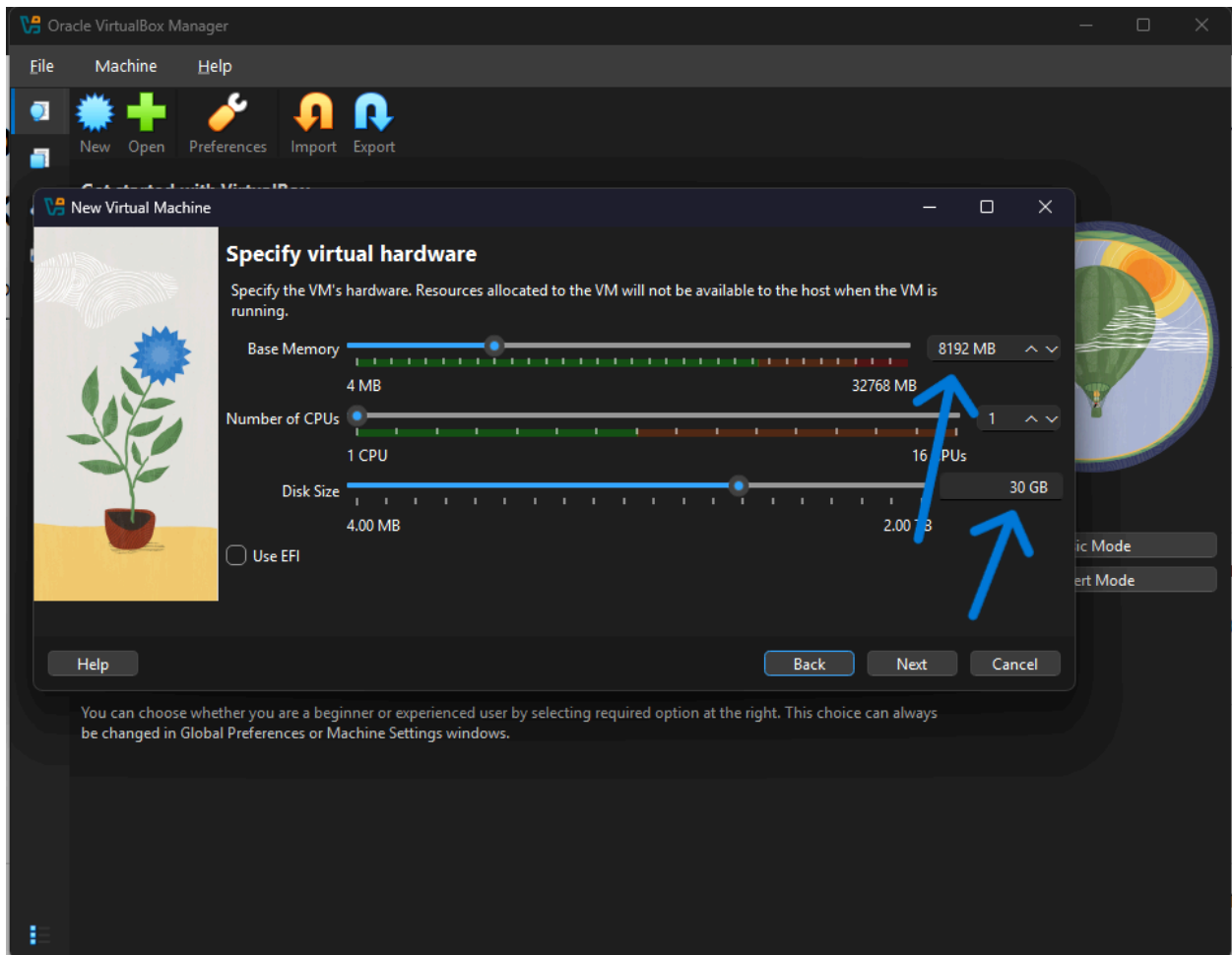


3. Once VirtualBox is installed and gets to the home screen, click new in the top left to get the configuration for our virtual machine started.



4. You will be greeted with a new page. This is where you can name your virtual machine. I called mine "Ubuntu-Lab." You can also attach the ISO file and configure the RAM and memory.
 - **NOTE:** If your system has 8GB RAM, you want to configure the system to 4096 MB minimum. If your system has more than 8GB RAM, then set the memory to 8192 MB.





Notes:

If you want, you can increase the memory and disk size. It is better to have at least 30GB of disk size as we will be installing tools and packages

Step 3 — Start the Virtual Machine

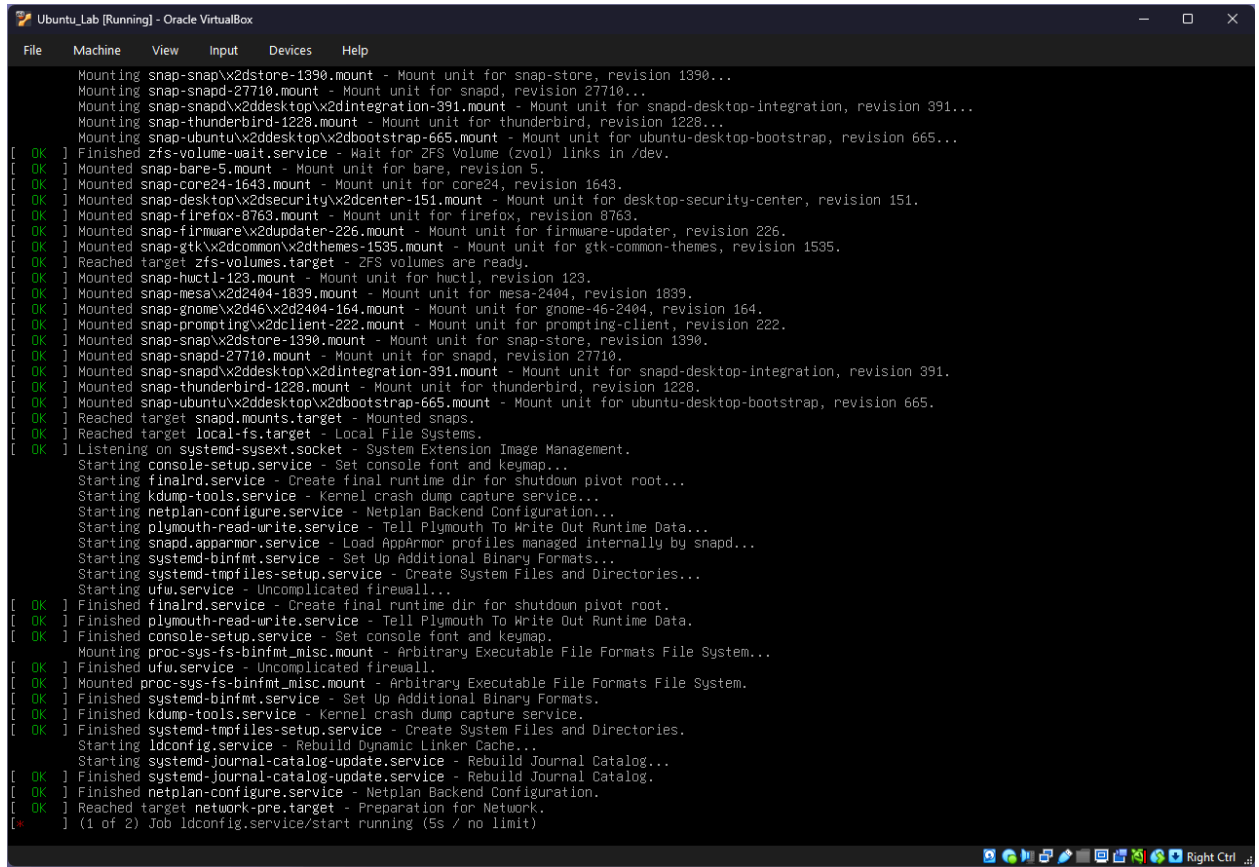
Objective

Start the newly created VM and begin the Ubuntu installation process.

Procedure

1. After everything is set up and configured, go ahead and run the virtual machine.

- The VM might take a minute to start, but once it does, install Ubuntu when the prompt is given and you will see a terminal pop up with the Ubuntu being installed



```
Ubuntu_Lab [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Mounting snap-snap\x2dstore-1390.mount - Mount unit for snap-store, revision 1390...
Mounting snap-snapd-27710.mount - Mount unit for snapd, revision 27710...
Mounting snap-snapd\x2ddesktop\x2dintegration-391.mount - Mount unit for snapd-desktop-integration, revision 391...
Mounting snap-thunderbird-1228.mount - Mount unit for thunderbird, revision 1228...
Mounting snap-ubuntu\x2ddesktop\x2dbootstrap-665.mount - Mount unit for ubuntu-desktop-bootstrap, revision 665...
[ OK ] Finished zfs-volume-wait.service - Wait for ZFS Volume (zvol) links in /dev.
[ OK ] Mounted snap-bare-5.mount - Mount unit for bare, revision 5.
[ OK ] Mounted snap-core24-1643.mount - Mount unit for core24, revision 1643.
[ OK ] Mounted snap-desktop\x2dsecurity\x2dcenter-151.mount - Mount unit for desktop-security-center, revision 151.
[ OK ] Mounted snap-firefox-8763.mount - Mount unit for firefox, revision 8763.
[ OK ] Mounted snap-firmware\x2dupdater-226.mount - Mount unit for firmware-updater, revision 226.
[ OK ] Mounted snap-gtk\x2dcommon\x2dthemes-1535.mount - Mount unit for gtk-common-themes, revision 1535.
[ OK ] Reached target zfs-volumes.target - ZFS volumes are ready.
[ OK ] Mounted snap-hucl1-123.mount - Mount unit for hucl1, revision 123.
[ OK ] Mounted snap-mesa\x2d2404-1839.mount - Mount unit for mesa-2404, revision 1839.
[ OK ] Mounted snap-gnome\x2d46\x2d2404-164.mount - Mount unit for gnome-46-2404, revision 164.
[ OK ] Mounted snap-prompting\x2dclient-222.mount - Mount unit for prompting-client, revision 222.
[ OK ] Mounted snap-snap\x2dstore-1390.mount - Mount unit for snap-store, revision 1390.
[ OK ] Mounted snap-snapd-27710.mount - Mount unit for snapd, revision 27710.
[ OK ] Mounted snap-snapd\x2ddesktop\x2dintegration-391.mount - Mount unit for snapd-desktop-integration, revision 391.
[ OK ] Mounted snap-thunderbird-1228.mount - Mount unit for thunderbird, revision 1228.
[ OK ] Mounted snap-ubuntu\x2ddesktop\x2dbootstrap-665.mount - Mount unit for ubuntu-desktop-bootstrap, revision 665.
[ OK ] Reached target snapd.mounts.target - Mounted snaps.
[ OK ] Reached target local-fs.target - Local File Systems.
[ OK ] Listening on systemd-sysexl.socket - System Extension Image Management.
Starting console-setup.service - Set console font and keymap...
Starting finalrd.service - Create final runtime dir for shutdown pivot root...
Starting kdump-tools.service - Kernel crash dump capture service...
Starting netplan-config.service - Netplan Backend Configuration...
Starting plymouth-read-write.service - Tell Plymouth To Write Out Runtime Data...
Starting snapd.apparmor.service - Load AppArmor profiles managed internally by snapd...
Starting systemd-binfmt.service - Set Up Additional Binary Formats...
Starting systemd-tmpfiles-setup.service - Create System Files and Directories...
Starting ufw.service - Uncomplicated firewall...
[ OK ] Finished finalrd.service - Create final runtime dir for shutdown pivot root.
[ OK ] Finished plymouth-read-write.service - Tell Plymouth To Write Out Runtime Data.
[ OK ] Finished console-setup.service - Set console font and keymap.
Mounting proc-sys-fs-binfmt_misc.mount - Arbitrary Executable File Formats File System...
[ OK ] Finished ufw.service - Uncomplicated firewall.
[ OK ] Mounted proc-sys-fs-binfmt_misc.mount - Arbitrary Executable File Formats File System.
[ OK ] Finished systemd-binfmt.service - Set Up Additional Binary Formats.
[ OK ] Finished kdump-tools.service - Kernel crash dump capture service.
[ OK ] Finished systemd-tmpfiles-setup.service - Create System Files and Directories.
Starting ldconfig.service - Rebuild Dynamic Linker Cache...
Starting systemd-journal-catalog-update.service - Rebuild Journal Catalog...
[ OK ] Finished systemd-journal-catalog-update.service - Rebuild Journal Catalog.
[ OK ] Finished netplan-config.service - Netplan Backend Configuration.
Reached target network-pre.target - Preparation for Network.
(1 of 2) Job ldconfig.service/start running (5s / no limit)
```

- After everything is installed, you will be brought to the home screen of your new UbuntuOS.

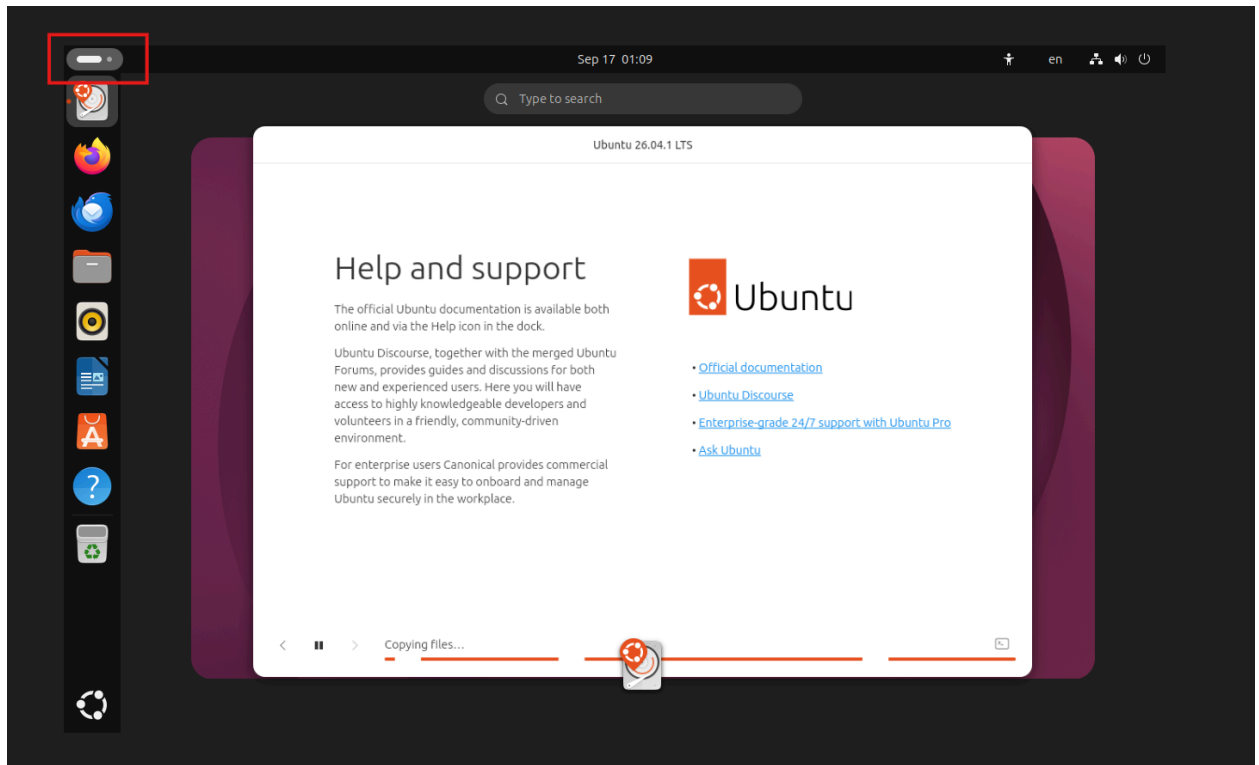
Step 4 — Finding and Opening Terminal

Objective

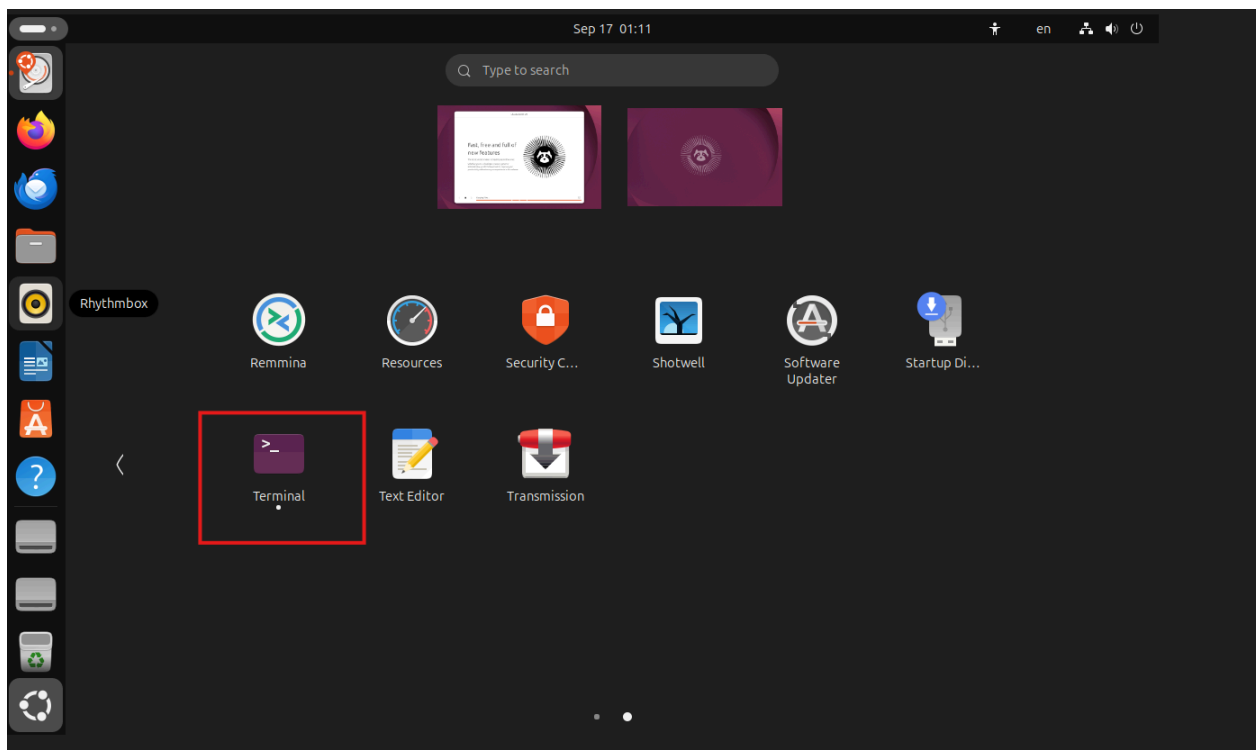
Finding and getting into Terminal.

Procedure

- There are going to be two dots on the top left of your Ubuntu home page. Click those dots to open up a vertical taskbar.



2. After the taskbar is open, you will see a Ubuntu logo at the bottom left. Click that to open the complete list of applications.
3. The terminal should be listed somewhere between all the applications



4. Double click on the terminal to open it up.

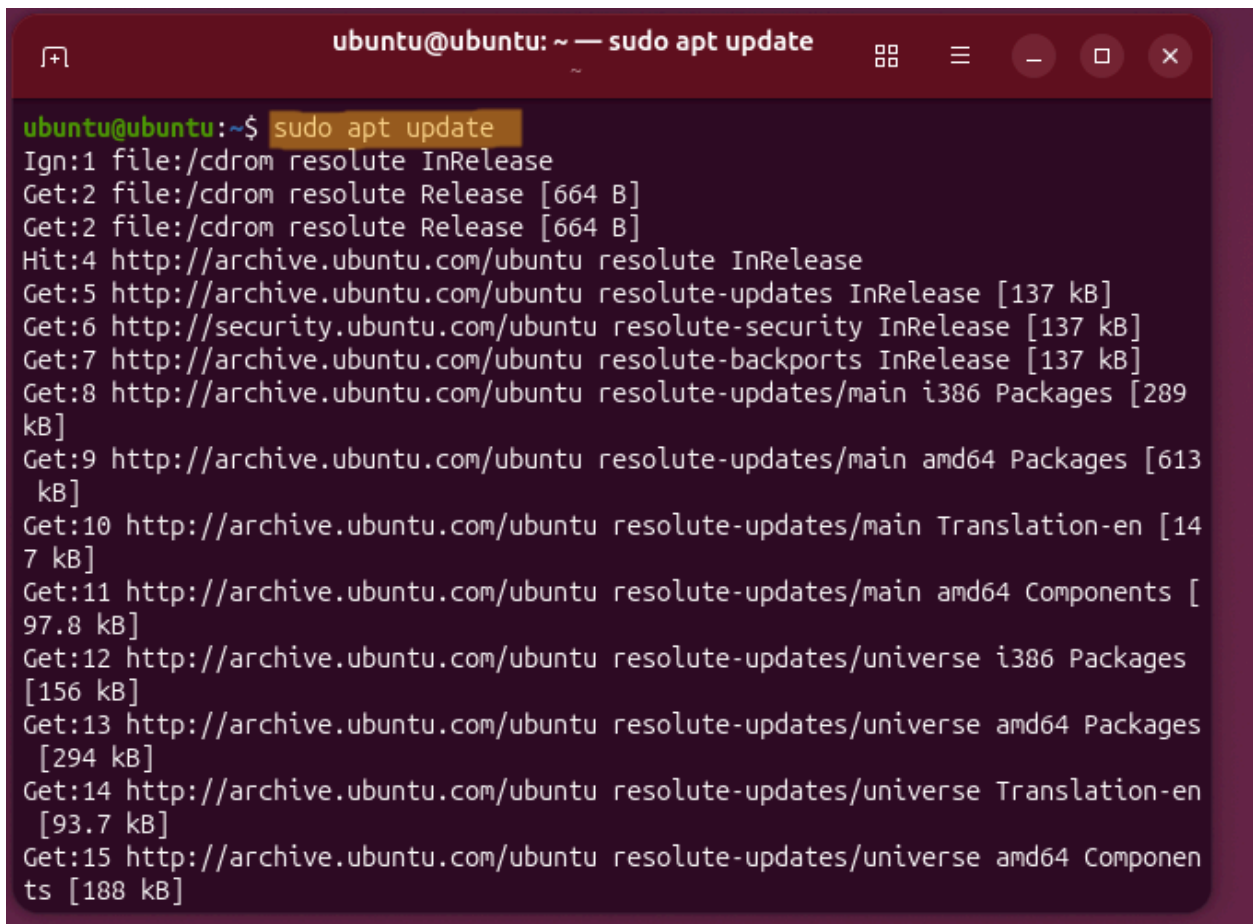
Step 5 — Running Terminal Commands and What They Mean

Objective

Run the basic and important commands in the Terminal, as well as understanding what each command does.

Procedure

1. `sudo apt update`
 - a. Checks for the latest package list



```
ubuntu@ubuntu: ~ — sudo apt update
ubuntu@ubuntu:~$ sudo apt update
Ign:1 file:/cdrom/resolute InRelease
Get:2 file:/cdrom/resolute Release [664 B]
Get:2 file:/cdrom/resolute Release [664 B]
Hit:4 http://archive.ubuntu.com/ubuntu/resolute InRelease
Get:5 http://archive.ubuntu.com/ubuntu/resolute-updates InRelease [137 kB]
Get:6 http://security.ubuntu.com/ubuntu/resolute-security InRelease [137 kB]
Get:7 http://archive.ubuntu.com/ubuntu/resolute-backports InRelease [137 kB]
Get:8 http://archive.ubuntu.com/ubuntu/resolute-updates/main i386 Packages [289 kB]
Get:9 http://archive.ubuntu.com/ubuntu/resolute-updates/main amd64 Packages [613 kB]
Get:10 http://archive.ubuntu.com/ubuntu/resolute-updates/main Translation-en [147 kB]
Get:11 http://archive.ubuntu.com/ubuntu/resolute-updates/main amd64 Components [97.8 kB]
Get:12 http://archive.ubuntu.com/ubuntu/resolute-updates/universe i386 Packages [156 kB]
Get:13 http://archive.ubuntu.com/ubuntu/resolute-updates/universe amd64 Packages [294 kB]
Get:14 http://archive.ubuntu.com/ubuntu/resolute-updates/universe Translation-en [93.7 kB]
Get:15 http://archive.ubuntu.com/ubuntu/resolute-updates/universe amd64 Components [188 kB]
```

- a. **git** = upload projects to GitHub
- b. **curl** / **wget** = download files
- c. **vim** = edit text files
- d. **unzip** = extract files
- e. **net-tools** = networking commands
- f. **ssh** = remote login
- g. **ansible** = automation
- h. **python3** & **pip** = scripting
- i. **pip** = allows the install of Python packages

```
utanvee1@Ubuntu-Lab:~$ sudo apt install -y git curl wget vim unzip net-tools ssh ansible python3 python3-pip
[sudo: authenticate] Password:
wget is already the newest version (1.25.0-2ubuntu4.4).
wget set to manually installed.
unzip is already the newest version (6.0-29ubuntu1).
unzip set to manually installed.
python3 is already the newest version (3.14.3-0ubuntu2).
python3 set to manually installed.
The following packages were automatically installed and are no longer required:
  linux-headers-7.0.0-30      linux-image-unsigned-7.0.0-30-generic  linux-modules-7.0.0-30-generic  linux-tools-7.0.0-30-generic
  linux-headers-7.0.0-30-generic  linux-main-modules-zfs-7.0.0-30-generic  linux-tools-7.0.0-30
Use 'sudo apt autoremove' to remove them.

Installing:
  ansible  curl  git  net-tools  python3-pip  ssh  vim

Installing dependencies:
  ansible-core  gcc-15-x86-64-linux-gnu  libexpat1-dev  libstdc++-15-dev  python3-dnspython  python3-worm  python3-winrm
  build-essential  gcc-x86-64-linux-gnu  libfakeroot  libtcl  python3-gssapi  python3-xmltodict
  bz2  git-man  libfile-fcntllock-perl  libubsan1  python3-jmespath  python3.14-dev
  dpkg-dev  libalgorithm-diff-perl  libgcc-15-dev  lto-disabled-list  python3-kerberos  ssh-import-id
  fakeroot  libalgorithm-diff-xs-perl  libhwasan0  make  python3-libcloud  vim-runtime
  g++  libalgorithm-merge-perl  libitm1  ncurses-term  python3-libcloud  zlib1g-dev
  g++-15  libasan8  libltdl  openssh-server  python3-packaging  python3-pyspnego
  g++-15-x86-64-linux-gnu  libbrotli  libpython3-dev  openssh-sftp-server  python3-requests-ntlm  python3-resolvlib
  g++-x86-64-linux-gnu  libcrypt-dev  libpython3.14-dev  python3-argcomplete  python3-resolvlib  python3-selinux
  gcc  libdpkg-perl  libquadmath0  python3-decorator  python3-selinux  python3-wheel
  gcc-15  liberror-perl  libsodium23  python3-dev  python3-wheel

Suggested packages:
  cowsay  g++-multilib  automake  gcc-15-multilib  git-gui  bzr  ssh-askpass  python3-trio
  sshpass  g++-15-multilib  libtool  gcc-15-locales  gitk  libstdc++-15-doc  python3-aiogram  ctags
  bz2-doc  gcc-15-doc  flex  gdb-x86-64-linux-gnu  gitweb  make-doc  python3-h2  vim-doc
  debconf  gcc-multilib  bison  git-doc  git-cvs  molly-guard  python3-httpcore  vim-essentials
```

NOTE: This step will ask for your password. Enter the password you originally used when setting up your VM to continue installing the tools

Step 6 — Enabling SSH

Objective

Check to see if SSH is installed and active. SSH is important as it is the most common Linux admin skill and is important for remote work in Linux.

Procedure

1. Open the Terminal.
2. `ssh -V`
 - a. Checks to see if SSH is installed
 - i. If not installed, run: `sudo apt install -y openssh-server`
3. `sudo systemctl status ssh`
 - a. Checks the SSH status
4. If the status is showing disabled, run:
 - a. `sudo systemctl enable ssh`

b. `sudo systemctl start ssh`

Step 7 — Take a Snapshot of VM

Objective

Create a snapshot of our virtual machine. A snapshot is a saved version of your virtual machine. This way, if anything breaks, you can go back to the “clean state.” This will be extremely useful when doing labs.

Procedure

1. In the top left of your VM terminal, you will see “file.”
 2. Click on file, and choose the snapshot option.
 3. The virtual machine will load for a second, and you will have taken a snapshot of your VM.
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5. Revision History

Version	Date	Author	Description
1.0	09.16.26	Uzair Tanveer	Initial SOP created

Appendix A — Commands Used

Document all Linux commands used throughout the setup.

`sudo apt update` — checks for latest packages list

`sudo apt upgrade -y` — installs updates automatically

`sudo apt install -y git curl wget vim unzip net-tools ssh ansible python3 python3-pip` — installs common tools that would be useful for projects

`ssh -V` — checks if SSH is installed

`sudo apt install -y openssh-server` — installs SSH

`sudo systemctl status ssh` — checks status of SSH

`sudo systemctl enable ssh` — enables SSH

`sudo systemctl start ssh` — starts SSH
